SUMMARY

The state of a gas is defined by a relationship between the four independent variables pressure (P), volume (V), temperature (T) and number of moles (n). The relationship between these parameters is governed by different gas laws as summarised below.

LAW	Expression	Conditions	Expression for two differnt states
Boyle's law	PV = constant	n and T are constant	$P_1V_1 = P_2V_2$
Charle's law	$\frac{V}{T}$ = constant	P and n are constant	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Gay Lussac's law	$\frac{P}{T}$ = constant	V and n are constant	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Combined gas law	$\frac{PV}{T}$ = constant	n is constant	$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$
Avogadro's Hypothesis	$\frac{V}{n}$ = constant	T and P are constant	$\frac{V_1}{n_1} = \frac{V_2}{n_2}$
Dalton's law of partial pressure	$P_{Total} = p_1+p_2+p_3+....$	T and V are constant	
Graham's law of diffusion	$r \alpha \frac{1}{\sqrt{M}}$	T and P are constant	$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$

Gases that obey the equation PV=nRT under all conditions are called ideal gases. But in practice there is no ideal gas. Gases tend to behave ideally at high temperatures and at low pressures. For real gases, van der Waals modified the ideal gas equation as

$$\left(P + \frac{a\,n^2}{V^2}\right)(V - nb) = nRT$$

Critical temperature (T_c) of a gas is defined as the temperature above which it cannot be liquefied at any pressure. Critical pressure (P_c) of a gas is defined as the minimum pressure required to liquefy 1 mole of a gas at its critical temperature. Critical volume (V_c) is defined as the volume occupied by 1 mole of a gas at its critical temperature and critical pressure.

11th Std Chemistry 158-185.indd 176 06-01-2021 15:35:25